# AZ-305 · Extras — Missing Topic Locks

Overflow facts · color = source domain: D1 blue · D2 green · D3 orange · D4 purple

## D4 VM family decoder

| Letter | Meaning | Trigger words |
|---|---|---|
| A | Ancient/basic | Cheap dev/test; **no AN** |
| B | Burstable | Intermittent CPU |
| D | Default | General-purpose |
| E/M | Extra/massive memory | DB, cache, analytics / SAP HANA |
| F | Fast CPU | Compute optimized |
| L | Local disk | Storage IOPS / local NVMe |
| N | NVIDIA GPU | NC compute · NV viz · ND deep learning |
| H | HPC | MPI, CFD, weather, FEA, InfiniBand |
| DC | Data confidential | TEE / secure enclave |

**Suffix: s**=Premium storage · **d**=local temp disk · **_v#**=generation.

## D4 disk SKU by IOPS

| Need | Pick |
|---|---|
| ≤500 IOPS, cheapest | **Standard HDD** |
| ≤6K IOPS (E80 max) | **Standard SSD** |
| Tiered ~20K (P30 5K, P40 7.5K, P60 16K) | **Premium SSD** |
| Exact IOPS low cost, size decoupled | **Premium SSD v2** |
| ≥20K IOPS or sub-ms latency | **Ultra Disk** |

## D2 SQL data protection

- Hide sensitive values at query time → **DDM**
- Encrypt whole DB at rest → **TDE**
- Encrypt column from DBAs → **Always Encrypted**
- Restrict who can query table → **SQL roles**
- Audit who queried what / wire → **Auditing / TLS**

DDM masks results (XXX-XX-1234); RBAC/SQL roles control access, not visibility. TDE is default ON for Azure SQL.

## D1 boundary + identity location

- Non-Azure infra into ARM → **Arc**
- Cross-tenant RBAC / MSP SSO → **Lighthouse**
- Cross-tenant data snapshot/in-place → **Data Share**
- Azure compute / Arc-enabled / other cloud → **MI / Arc MI / SP**

Real person = user + CA/MFA/PIM. On-prem no Arc or other cloud = SP (cert > secret). Never PATs for app auth.

## D3 Key Vault DR trio

- TDE protector key type/sizes → **RSA 2048/3072/4096**
- Never for TDE protector → **AES**
- Auto-failover crypto ops → **Work**
- Auto-failover writes → **Blocked**
- Customer backup/restore → **Same geography***

Failover = MS-managed paired region, read-only. Get/Wrap/Unwrap/Encrypt/Decrypt work; Set/Create/Delete blocked. *Brazil South ↔ South Central US is the cross-geo backup/restore exception.

## D3 DB HA / DR tier traps

| Requirement | Pick |
|---|---|
| MySQL/PostgreSQL zone outage | **Zone-redundant HA** |
| Region fails, min cost | **Geo-redundant backup** |
| Lower RTO/RPO regional DR | **Cross-region read replica** |
| SQL DB DTU zone redundant | **Premium only** |
| SQL DB vCore zone redundant | **GP / BC / Hyperscale** |

Exam trap: cheapest region DR for Flexible Server = geo-backup restore, not read replica. BC has sync replicas / zero data loss for AZ failover.

## D1 app registration + Graph

| Blade / suffix | Meaning |
|---|---|
| App roles | Roles I offer to callers (API/server) |
| API permissions | Perms I want to call (client) |
| Expose an API | Delegated scopes + authorized clients |
| Token config | Optional claims |
| Enterprise app assignment required | Yes + assign SPs to app role |
| No suffix / .Shared / .All | Own data / shared with user / tenant-wide |

**Application permissions** require admin consent only; no user consent. Use assignment required when only specific service principals may call.

## D2 SAS + Files auth

- Shared key → **Avoid**
- Account SAS / Service SAS → **Key-signed**
- Revocable Service SAS → **Stored access policy**
- Max security, no key, time-limited → **User delegation SAS**
- Azure Files SMB modern / legacy → **Entra Kerberos / AD DS**

Signer RBAC: Blob Data Reader = read SAS; Contributor = write SAS; Owner = ACL. Entra Domain Services also legacy-ish for SMB.




## D1 Log Analytics cost levers

| Requirement | Pick |
|---|---|
| Hot query default 30d→2y (~$2.30/GB) | **Analytics/Interactive** |
| Must ingest all + minimize cost (~75% less) | **Basic Logs** |
| Even cheaper / long retention up to 12y | **Auxiliary / Archive** |
| Known daily volume discount | **Commitment tiers** |
| Drop/edit before ingest | **DCR transform** |
| Query Archive / hard drop ceiling | **Search job or Restore / cap** |

Basic Logs: cheaper ingest, pay-per-query; Simple Log Alerts supported (≥1-min freq; 100/5K limit for 1-min rules). Archive ≈$0.02/GB-mo; query via Search jobs or Restore.

## D4 SQL migration + moves

- AG-source → SQL MI bridge → **MI Link**
- AG-source / cross-OS seconds cutover → **Distributed AG**
- Single SQL online to SQL DB/MI/VM → **DMS online**
- Post-migration SQL MI cross-region DR → **Auto-failover group**
- Small DBs / same-region / cross-region / DR only → **Backup URL / move / RM / ASR**

MI Link uses Distributed AG under the hood. Auto-failover group: Microsoft-managed, planned = no data loss, unplanned may lose recent writes.

## D4 AKS scaling

- More pods in existing deployment → **HPA**
- More nodes in a node pool → **Cluster Autoscaler**
- Bursty Linux stateless pods → **Virtual Nodes**
- Event-driven scale-to-zero → **KEDA**

HPA scales pods; Cluster Autoscaler scales nodes; Virtual Nodes burst to ACI (Linux only).

## D4 App Service plan vs ASE

- Normal web app hosting / shared platform → **App Service plan**
- Need private inbound only → **Private Endpoint**
- Single-tenant dedicated VNet env → **ASE v3**
- >30 instances / high scale / compliance isolation → **ASE v3**

Plan = compute profile for apps; apps in same plan share workers. ASE = isolated App Service stamp in your subnet; uses Isolated v2 plans.